

Week 9: Bayesian Computations

MATH-517 Statistical Computation and Visualization

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Section 1

Bayesian Inference

Bayes' Rule

Let X be a random variable and θ a parameter, considered also a random variable:

$$f_{X,\theta}(x, \theta) = \underbrace{f_{X|\theta}(x | \theta)}_{\text{likelihood}} \underbrace{f_{\theta}(\theta)}_{\text{prior}} = \underbrace{f_{\theta|X}(\theta | x)}_{\text{posterior}} f_X(x).$$

- likelihood = frequentist model (θ fixed)
- likelihood & prior = Bayesian model (θ random)

Denoting by x_0 the observed value of X :

$$f_{\theta|X=x_0}(\theta | x_0) = \frac{f_{X|\theta}(x_0 | \theta) f_{\theta}(\theta)}{f_X(x_0)} = \frac{f_{X|\theta}(x_0 | \theta) f_{\theta}(\theta)}{\int f_{X|\theta}(x_0 | \theta) f_{\theta}(\theta) d\theta},$$

which is the Bayes' rule. Rewritten:

$$f_{\theta|X=x_0}(\theta | x_0) \propto f_{X|\theta}(x_0 | \theta) f_{\theta}(\theta),$$

in words: posterior $\propto$ likelihood $\times$ prior

$\propto$... proportional to

Information update

$X = x_0$ and/or $\theta = (\psi, \lambda)$ can even be vectors:

$$f_{\psi|X=x_0}(\psi | x_0) \propto \int f_{X|\theta}(x_0 | \theta) f_{\theta}(\psi, \lambda) d\lambda$$

- our original (prior) information (belief) about θ was updated by observing $X = x_0$ into the (marginal) posterior
- this can be applied recursively (when a new Y independent of X arrives):

$$\begin{aligned} f_{\theta|X=x, Y=y}(\theta | x_0, y_0) &\propto f_{Y, X|\theta}(x_0, y_0 | \theta) f_{\theta}(\theta) \\ &= f_{Y|\theta}(y_0 | \theta) \underbrace{f_{X|\theta}(x_0 | \theta) f_{\theta}(\theta)}_{\text{old posterior}}, \end{aligned}$$

All available information about θ is summarized by the posterior (provides a complete inferential scope)

Prior Densities

- the prior distribution quantifies the researcher's uncertainty about parameters before observing data
- choice of the prior density is important: it is based on the best available information $\rightarrow$ subjective
- sometimes use an **improper** prior, which is not a true density (has infinite integral) but for which the posterior is a true density
- Watch out when using improper priors, there is no guarantee that the posterior distribution will exist!
- **conjugate** priors make computations easy as they yield a posterior density of the same family (e.g., beta prior/binomial data $\rightarrow$ beta posterior or gamma prior/Poisson data $\rightarrow$ gamma posterior)

Prior Densities

- often use a **non-informative** proper prior, which inputs only **weak information** ($\neq$ ignorance) (e.g., normal density with very large (finite) variance or uniform prior though not transformation-invariant)
- two main types of non-informative priors:
 - Laplace prior for which $\pi(\theta) \propto 1_{\{\theta \in \Theta\}}$ (might be improper)
 - Jeffreys' prior for which

$$\pi(\theta) \propto \sqrt{\det I(\theta)}$$

where for any (non random) $\theta \in \Theta$, $I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} \ln f(X; \theta) \right]$
with $X \sim f(\cdot; \theta)$

Note: Empirical Bayes can be used to select parameters of the prior (approximate prior distribution by frequentist methods). Other techniques: hierarchical Bayes or maximum entropy

Prior Densities

- as sample size increases, the effect of the prior is washed out

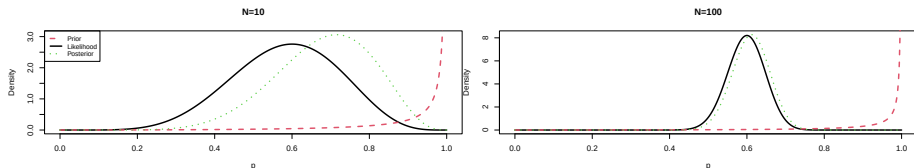
E.g., Bernoulli case

- likelihood: $\Pr(\mathbf{x}_{1:N}|p) = p^{\sum_{i=1}^N x_i} (1-p)^{N-\sum_{i=1}^N x_i}$
- beta prior $p \sim \text{Beta}(a, b)$:

$$\Pr(p) \propto p^{a-1} (1-p)^{b-1}$$

on the interval $(0, 1)$

- posterior: $\Pr(p|\mathbf{x}_{1:N}) \propto \Pr(\mathbf{x}_{1:N}|p) \Pr(p)$



Prior Densities: Example

- **Improper Prior**

If $x \sim \mathcal{N}(\theta, 1)$ and $f_\theta(\cdot) \equiv \omega$ (constant), then posterior dist. of θ is

$$f_{\theta|x}(\theta | x) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(x - \theta)^2}{2} \right\},$$

i.e., corresponds to a $\mathcal{N}(x, 1)$ distribution $\Rightarrow$ independent of the prior

- **Non-informative (flat) prior**

Consider $x \sim \mathcal{N}(\theta, 1)$ and $\theta \sim \mathcal{N}(0, 10)$

$$\begin{aligned} f_{\theta|x}(\theta | x) &\propto f(x | \theta) f_\theta(\theta) \propto \exp \left\{ -\frac{(x - \theta)^2}{2} - \frac{\theta^2}{20} \right\} \\ &\propto \exp \left(-\frac{11\theta^2}{20} + \theta x \right) \propto \exp \left[-\frac{11}{20} \left\{ \theta - (10x/11) \right\}^2 \right] \end{aligned}$$

and

$$\theta | x \sim \mathcal{N} \left(\frac{10}{11}x, \frac{10}{11} \right)$$

The Bayesian Approach

- let us denote the data set D , its realization d , and θ the parameter(s)
- the Bayesian model assumes
 - that nature picks θ from the prior distribution f_θ
 - that nature generates data set $D = d$ from the likelihood $f_{D|\theta}$
- the posterior

$$f(\theta \mid D = d) \propto f(d \mid \theta)f(\theta)$$

provides answers for all statistical tasks

- point estimation
- interval estimation
- prediction
- model selection
- hypothesis testing? a matter of choosing priors reflecting the hypotheses
- **Uncertainty:**
 - How much prior belief about θ changes in light of data (Bayesian)
 - How estimates vary in repeated sampling from the same population (frequentist)

Point estimation

Goal: a numerical value $\hat{\theta}$ compatible with the data

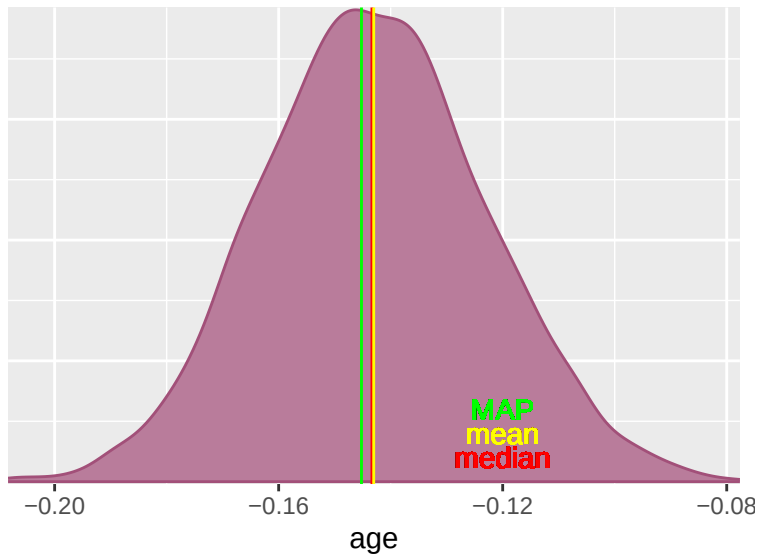
Frequentist approach:

- MLE
- method of moments
- optimization (e.g. penalized least squares), etc.

Bayesian approach:

- MAP - Maximum A Posterior estimate
 - the maximum of the posterior density (close to frequentist MLE)
- posterior mean - the expected value of the posterior
- posterior median
- generally: minimizing the expected loss
 - the expectation is calculated under the posterior
 - e.g., for the squared error loss $L(\theta, a) = (\theta - a)^2$, the posterior expected loss $\int_{\Theta} (\theta - a)^2 dF_{\theta|D}(\theta)$ is minimized at the mean of the posterior distribution; $|\theta - a|$ yields the posterior median

Point Estimation



Interval Estimation

Goal: a range of values $\hat{\theta}$ compatible with the data

Frequentist approach: a confidence interval $CI_{1-\alpha}$

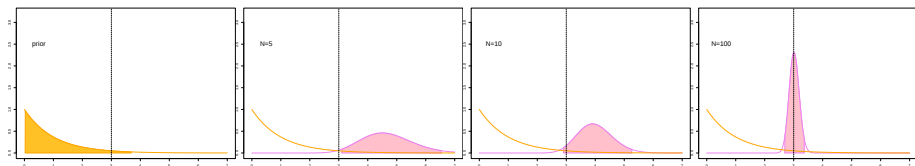
- dual to significance testing
- the probability that the interval contains the true parameter *under replication of the data* is $1 - \alpha$

Bayesian approach: a credible set $CR_{1-\alpha}$

- a subset of Θ such that $P(\theta \in CR_{1-\alpha} \mid D) = 1 - \alpha$
 - probability calculated under the posterior
- simple interpretation: given the model and data, the probability that the true parameter is in the credible set is $1 - \alpha$
- Infinitely many such intervals/regions
- many options (just as in the frequentist context), most used: equal-tailed interval and the *highest posterior density set* (narrowest possible)

Interval Estimation: Equal-tailed Interval

- $CR_{1-\alpha} = [q_{\alpha/2}, q_{1-\alpha/2}]$ where q_{α} is the α -quantile of the posterior distribution $f_{\theta|D}$



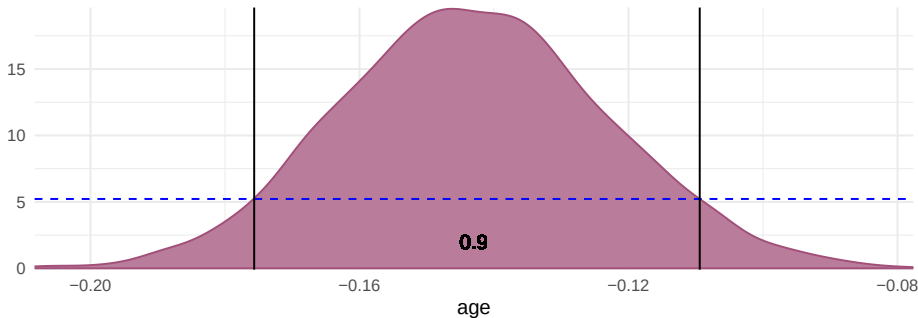
- credible interval influenced by the prior
- credible interval gets narrower with increasing N
- may include values with lower probability than those excluded, unless the posterior is unimodal and symmetric

Interval Estimation: Highest Posterior Density Set

- $\int_{\Theta \cap CR_{1-\alpha}} f_{\theta|D}(\theta | d) d\theta = 1 - \alpha$ such that

$$f_{\theta|D}(\theta | d) \geq f_{\theta|D}(\theta' | d)$$

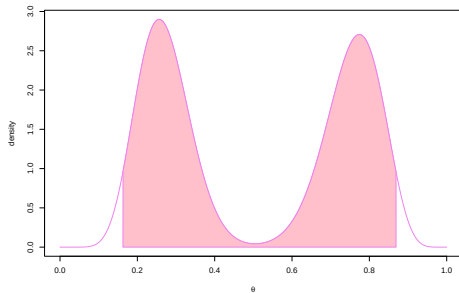
for all $\theta \in CR_{1-\alpha}$ and $\theta' \notin CR_{1-\alpha}$



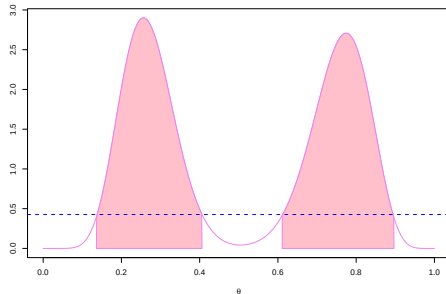
- not necessarily an interval: if the posterior is multimodal, the HPD set may be an union of distinct intervals (or distinct contiguous regions)

Interval Estimation: Bimodal Example

Equal-tailed 95% CR



95% HPD CR



Interval Estimation: Example

- Data $X_i \mid \mu, \tau \sim N(\mu, \tau^{-1}), i = 1, \dots, N$. Suppose τ is known, and that we use prior $\mu \sim N(\mu_0, \tau_0^{-1})$ for some fixed values of μ_0 and $\tau_0 \geq 0$
- The corresponding posterior distribution of μ is

$$\mu \mid x \sim N(\mu_p, \tau_p^{-1})$$

where $\tau_p = N\tau + \tau_0$ and $\mu_p = \frac{N\tau}{\tau_0 + N\tau}\bar{x} + \frac{\tau_0}{\tau_0 + N\tau}\mu_0$

Hence, 95% credible interval (in this case also 95% HPD region):

$$[\mu_p - z_{0.025}\sigma_p, \mu_p + z_{0.025}\sigma_p]$$

$\Rightarrow$ a priori info about a parameter decreases our (posterior) uncertainty about it

Note that the credible interval corresponding to the noninformative prior $[\bar{x} - z_{0.025}\sigma/\sqrt{N}, \bar{x} + z_{0.025}\sigma/\sqrt{N}]$

coincides with the classical (frequentist) confidence interval

Prediction of Future Observations

Goal: posterior prediction, i.e., evaluating or sampling from the posterior predictive distribution $f_{\tilde{D}|D}$, where D is observed data and $\tilde{D}$ is yet to be observed data

Bayesian approach: prediction = estimation

- assume that likelihood satisfies $f_{D,\tilde{D}|\theta} = f_{D|\theta} \cdot f_{\tilde{D}|\theta}$, i.e., new and old data are independent given parameters
- then

$$\begin{aligned} f_{\theta,D,\tilde{D}} &= f_{D|\tilde{D},\theta} \cdot f_{\tilde{D},\theta} = f_{D|\theta} \cdot f_{\tilde{D}|\theta} \cdot f_{\theta} \\ &= f_{\theta,\tilde{D}|D} \cdot f_D \end{aligned}$$

- joint posterior: $f_{\theta,\tilde{D}|D} = f_{\tilde{D}|\theta} \cdot f_{\theta|D} \Rightarrow$ marginalize out θ

$$f_{\tilde{D}|D}(\tilde{d} | d) = \int_{\Theta} f_{\tilde{D}|\theta}(\tilde{d} | \theta) \cdot f_{\theta|D}(\theta | d) d\theta$$

→ estimated by MC if we can draw from posterior $f_{\theta|D}$

Model Selection

Consider a discrete set $\mathcal{M}$ of candidate models indexed by M (a parameter)

Goal: decide which candidate model fits best the data

E.g., $\mathcal{M}$ can be a mixture of K Gaussians (K is discrete random variable)

Frequentist approach: hypothesis testing, e.g., LRT

Bayesian approach: model selection = estimation (again)

- the data generation process is assumed to have additional level
 - the nature generates a model $M \in \mathcal{M}$ based on a prior f_M
 - then it generates θ conditionally on the model from $f_{\theta|M}$
 - finally the data are generated conditionally on the model and parameters from $f_{D|M,\theta}$
- calculate the posterior (now hierarchical):

$$\begin{aligned} f_{D,\theta,M} &= f_{D|\theta,M} \cdot f_{\theta,M} = f_{D|\theta,M} \cdot f_{\theta|M} \cdot f_M \\ &= f_{\theta,M|D} \cdot f_D \end{aligned}$$

posterior: $f_{\theta,M|D} \propto f_{D|\theta,M} \cdot f_{\theta|M} \cdot f_M$

... marginalize out θ again

Example: Bayesian Ridge

Consider a Gaussian linear model $Y = \mathbf{X}\beta + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma^2 I_{N \times N})$.

Consider the following priors:

- $\beta \sim \mathcal{N}(0, \tau^2 I_{p \times p})$
 - τ^2 is a hyperparameter - either fixed or with some hyperprior f_{τ^2}
- $f_{\sigma^2} \propto 1/\sigma^2$ (improper prior)

Then the posterior for $\theta = (\beta, \sigma^2, \tau^2)^\top$ is given by

$$f_{\theta|\mathbf{X}, Y}(\beta, \sigma^2, \tau^2 \mid \mathbf{X}, Y) \propto \frac{1}{\sigma^N} e^{-\frac{1}{2\sigma^2} (Y - \mathbf{X}\beta)^\top (Y - \mathbf{X}\beta)} \frac{1}{\tau^p} e^{-\frac{1}{2\tau^2} \beta^\top \beta} \frac{1}{\sigma^2} f_{\tau^2}(\tau^2)$$

Interestingly, the log-posterior for β is

$$\log f_{\dots}(\beta \mid \mathbf{X}, Y, \sigma^2, \tau^2) \propto -\frac{1}{2\sigma^2} (Y - \mathbf{X}\beta)^\top (Y - \mathbf{X}\beta) - \frac{1}{2\tau^2} \beta^\top \beta$$

so MAP here gives the ridge estimator for $\lambda = \sigma^2/\tau^2$

Computational Difficulty

The Bayesian approach above is

- conceptually straightforward and holistic, but
- in practice requires computationally demanding integration
 - the normalization constant
 - marginalization
 - calculating expectations

Possible solutions:

- analytic approximations to the posterior (e.g., Laplace)
- Monte Carlo
 - but the MC techniques we saw already are useful mostly in low-dimensional problems
 - Markov Chain Monte Carlo (MCMC): explore the space in a dependent way, focusing on the important regions

Section 2

Markov Chain Monte Carlo (MCMC)

MC versus MCMC

Goal: calculate $\mathbb{E}g(X)$ for some function g and random variable X

Monte Carlo (MC):

- draw independently $X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} X$
- approximate $\mathbb{E}g(X)$ empirically by $N^{-1} \sum_n g(X_n)$
 - works due to LLN

Markov Chain Monte Carlo (MCMC):

- draw $X^{(1)}, X^{(2)}, \dots, X^{(T)}$ as an *ergodic* Markov Chain on a state space $\mathcal{X}$ with *stationary distribution* equal to that of X
- approximate $\mathbb{E}g(X)$ empirically by $T^{-1} \sum_t g(X^{(t)})$
 - works due to the *ergodic theorem* (LLN for Markov sequences)

$$\frac{1}{T} \sum_t g(X_t) \xrightarrow[T \rightarrow \infty]{\text{a.s.}} \mathbb{E}g(X),$$

for any bounded function $g : \mathcal{X} \rightarrow \mathbb{R}$

Markov Chains

Definition (informal): A sequence of random variables $\{X^{(t)}\}_{t \geq 0}$ with values in $\mathcal{X} \subset \mathbb{R}^p$ such that

$$X^{(t+1)} \mid X^{(t)}, X^{(t-1)}, \dots, X^{(0)} \sim X^{(t+1)} \mid X^{(t)}$$

is called a discrete-time *Markov chain*

- the conditional distribution $X^{(t+1)} \mid X^{(t)}$ is given by the *transition kernel* $k(x, y)$
 - for $X^{(t)} = x$, the cond. density of $X^{(t+1)}$ is $k_x(y) := k(x, y)$
 - k has to meet some conditions on measurability and integrability
 - a Markov chain is fully determined by the transition kernel and the initial distribution!
- a distribution f is called the stationary distribution of a Markov chain associated with a transition kernel k if $fK = f$, i.e.,

$$\int_{\mathcal{X}} k(x, y) f(x) dx = f(y)$$

If $f_{t+1}(y) = \int k(x, y) f_t(x) dx = f_t(y)$, then we stay in the distribution f_t forever (stationary=invariant under time evolution)

Detailed Balance

Claim: If the following *detailed balance condition* holds

$$k(x, y)f(x) = k(y, x)f(y)$$

for a distribution f and a transition kernel k , then f is a stationary distribution of the MC associated with k (just integrate both sides over $\mathcal{X}$)

- k specifies the amount of flow between the points in the domain $\mathcal{X}$
- detailed balance: the forward flow $x \rightsquigarrow y$ is equal to the backward flow $y \rightsquigarrow x$
- equilibrium distribution is preserved: if $x \sim f$ before a transition, then this is also true afterwards
- let f_t denote the marginal distribution of $X^{(t)}$
 - f_0 is the initial distribution
 - the update $f_t \rightsquigarrow f_{t+1}$ is governed by k
 - no update $\Leftrightarrow f_t$ is the stationary distribution f
 - if $f_0 = f$, there will never be an update ... $f_t = f$ for all t

Definitions:

- A chain verifying the detailed balance condition (time-reversibility) is called *invariant*
- A chain is called *irreducible* if any point can be reached (using the kernel) starting from anywhere else

$$\forall u, v \in \mathcal{X}, \quad \exists t \text{ s.t. } P(X^{(t)} = u \mid X^{(0)} = v) > 0$$

Result: An irreducible and aperiodic Markov chain converges to a unique distribution, called stationary distribution

- An irreducible, aperiodic, and invariant chain is called *ergodic*

Theorem: If a chain is ergodic then its unique stationary distribution is the invariant distribution and $f_t \rightarrow f$ for $t \rightarrow \infty$ regardless of f_0

Running Monte Carlo via Markov Chains

Goal: For an arbitrary starting value $X^{(0)}$, construct a chain with a pre-specified stationary distribution f , typically the posterior $f_{\theta|D=d}$

- chain = function that generates $X^{(t+1)}$ depending on $X^{(t)}$
 - the transition kernel k is in the background
- produce a dependent sample $X^{(T_0)}, X^{(T_0+1)}, \dots$, marginally generated from f , sufficient for most approximation purposes

MCMC is more widely applicable than MC, but what about *mixing*?

- we initialize our chain from $f_0 \neq f$
 - because if we could draw from f , we would be doing MC instead
 - need to ensure irreducibility
- after a *while* $f_{T_0} \approx f$ so we have our first draw $X^{(T_0)} \sim f$
 - discard $X^{(0)}, \dots, X^{(T_0-1)}$ and continue the chain (now stationary)
 - need to ensure invariance (detailed-balance)

Problem: How to build a Markov chain with a given stationary distribution? We will see some recipes

Metropolis–Hastings

Idea: construct a candidate new value y by drawing from arbitrary conditional density $q(y \mid x)$ (called *proposal* distribution)

- detailed balance requires the right amount of flow between all $x, y \in \mathcal{X}$
- if there is too much flow $x \rightsquigarrow y$, re-map some part of it as $x \rightsquigarrow x$

Metropolis–Hastings (MH) algorithm:

- **Input:** a proposal density $q(y \mid x)$, the target f (up to a constant)
- **for** $t = 1, 2, \dots$, update $X^{(t-1)}$ to $X^{(t)}$ by
 - generate $U^{(t)} \sim q(\cdot \mid X^{(t-1)})$
 - define

$$\alpha(X^{(t-1)}, U^{(t)}) = \min \left\{ 1, \frac{f(U^{(t)})q(X^{(t-1)} \mid U^{(t)})}{f(X^{(t-1)})q(U^{(t)} \mid X^{(t-1)})} \right\}$$

- set $X^{(t)} := U^{(t)}$ with probability $\alpha(X^{(t-1)}, U^{(t)})$
- otherwise set $X^{(t)} := X^{(t-1)}$

(if the proposal is symmetric, q vanishes from the formula above)

MH: Convergence Properties

- MH Markov Chain satisfies the detailed balance condition with

$$k(y, x) = \alpha(x, y) q(y | x) + \int \{1 - \alpha(x, \xi)\} q(\xi | x) d\xi \delta_x(y)$$

where δ is the Dirac mass

- If $q(y | x) > 0, \forall x, y$, then the chain is irreducible
- If

$$P\left(\frac{f(U^{(t)})q(X^{(t-1)} | U^{(t)})}{f(X^{(t-1)})q(U^{(t)} | X^{(t-1)})} \geq 1\right) < 1,$$

that is the event $\{X^{(t+1)} = X^{(t)}\}$ is possible, then the chain is aperiodic

Thus, under the above two conditions, we have

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T g(X^{(t)}) = \int g(x) df(x), \quad a.s.$$

for $\mathbb{E}_f |g(X)| < \infty$

MH with Random Walk

- use a local perturbation as proposal

$$U^{(t)} = X^{(t-1)} + \epsilon_t,$$

where $\epsilon_t \sim g$, independent of $X^{(t-1)}$

- proposal g is a symmetric density centered at the current sample $X^{(t-1)}$
 - e.g., g is $\mathcal{N}(0, \sigma^2)$ hence $U^{(t)} \sim \mathcal{N}(X^{(t-1)}, \sigma^2)$
 - e.g., g is $\mathcal{U}[-\delta, \delta]$ hence $U^{(t)} \sim \mathcal{U}[X^{(t-1)} - \delta, X^{(t-1)} + \delta]$
- then,

$$\alpha(X^{(t-1)}, U^{(t)}) = \min \left(1, \frac{f(U^{(t)})}{f(X^{(t-1)})} \right)$$

MH with Random Walk

Verifying detailed balance is relatively simple in this case:

- detailed balance: $k(x, u)f(x) = k(u, x)f(u)$ for $x \rightsquigarrow u$
- $k(x, u)$ is given implicitly as the mixture of
 - moving away $x \rightsquigarrow u$ with probability $\alpha(x, u) = \min \{1, f(u)/f(x)\}$
 - u is drawn from $q(u \mid x)$ a symmetric density around x
 - equal to
$$k(x, u) = \alpha(x, u)q(u \mid x) = \alpha(x, u)q(u - x) = \alpha(x, u)q(x - u)$$
 - staying at x with probability $1 - \alpha(x, u)$
 - i.e., $x = u$... detailed balance trivially satisfied
- detailed balance: $\cancel{q(u \mid x)}\alpha(x, u)f(x) = \cancel{q(x \mid u)}\alpha(u, x)f(u)$
- this is trivial since for $f(x) \neq f(u)$ it is
 - either $\alpha(u, x) = 1$ and $\alpha(x, u) = f(u)/f(x)$ leading to

$$\frac{f(u)}{f(x)}f(x) = f(u)$$

- or the other way around

MH Remarks

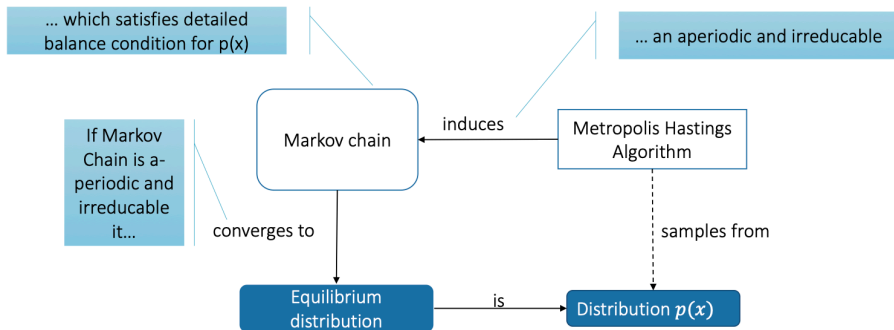
- f is usually a posterior, evaluations needed *up to normalization*
- MH similar in flavor to rejection sampling in MC
 - but RS needs a majorizing proposal g to decide accept vs. reject
 - MCMC instead moves vs. stays $\Rightarrow$ no majorization needed
- never moves to values with $f(y) = 0$
- the chain $(X^{(t)})_t$ may take the same value several times in a row, even when f is a density wrt Lebesgue measure

Def.: acceptance rate for MH is the average acceptance probability

$$\bar{\alpha} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \alpha(X^{(t-1)}, U^{(t)})$$

- if $\bar{\alpha}$ too large, we are probably not exploring the space, mostly staying close with our proposals to where we already were
- if $\bar{\alpha}$ too small, we have a lot of repeated values in our sample and hence the effective sample size is small even for large T
- good rates: 25% (large dimension) - 50% (small dimension)

MH: Summary



credit: Marcel Lüthi, University of Basel

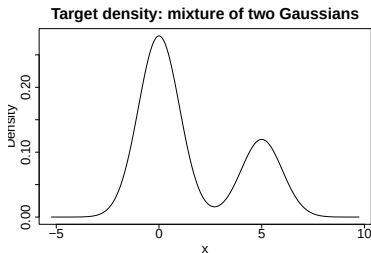
Example: MH with Random Walk

Consider the MH algorithm with

- a Gaussian mixture target model

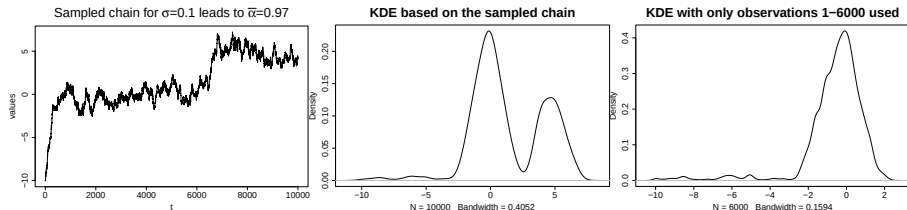
$$f_{\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \tau}(x) = \tau \varphi_{\mu_1, \sigma_1^2}(x) + (1 - \tau) \varphi_{\mu_2, \sigma_2^2}(x)$$

with $\mu_1 = 1, \mu_2 = 5, \sigma_1 = \sigma_2 = 1$ and $\tau = 0.7$

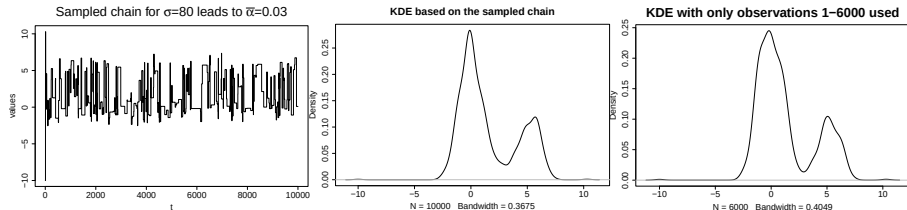


- a Gaussian random walk proposal $y \sim \mathcal{N}(x, \sigma^2)$, with $\sigma = 0.1, 3, 80$
- $x^{(0)} = -10$

Example: MH with Random Walk

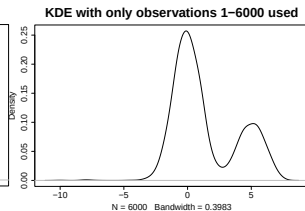
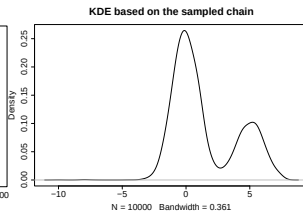
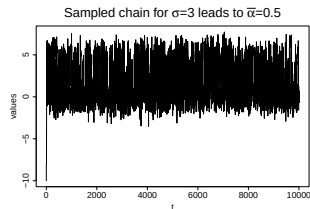


⇒ proposals often accepted but chain moves too slowly



⇒ chain gets stuck for too long

Example: MH with Random Walk



⇒ seems better

References

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C. P. Robert & G. Casella (2010) Introducing Monte Carlo Methods with *R*

Gelman & Carlin & Stern & Dunson & Vehtari & Rubin. 2013. Bayesian Data Analysis [updated version](#)